PORTFOLIO 2026

Chris Al-Rahi

Mechanical Engineering & Computing Technology at uOttawa

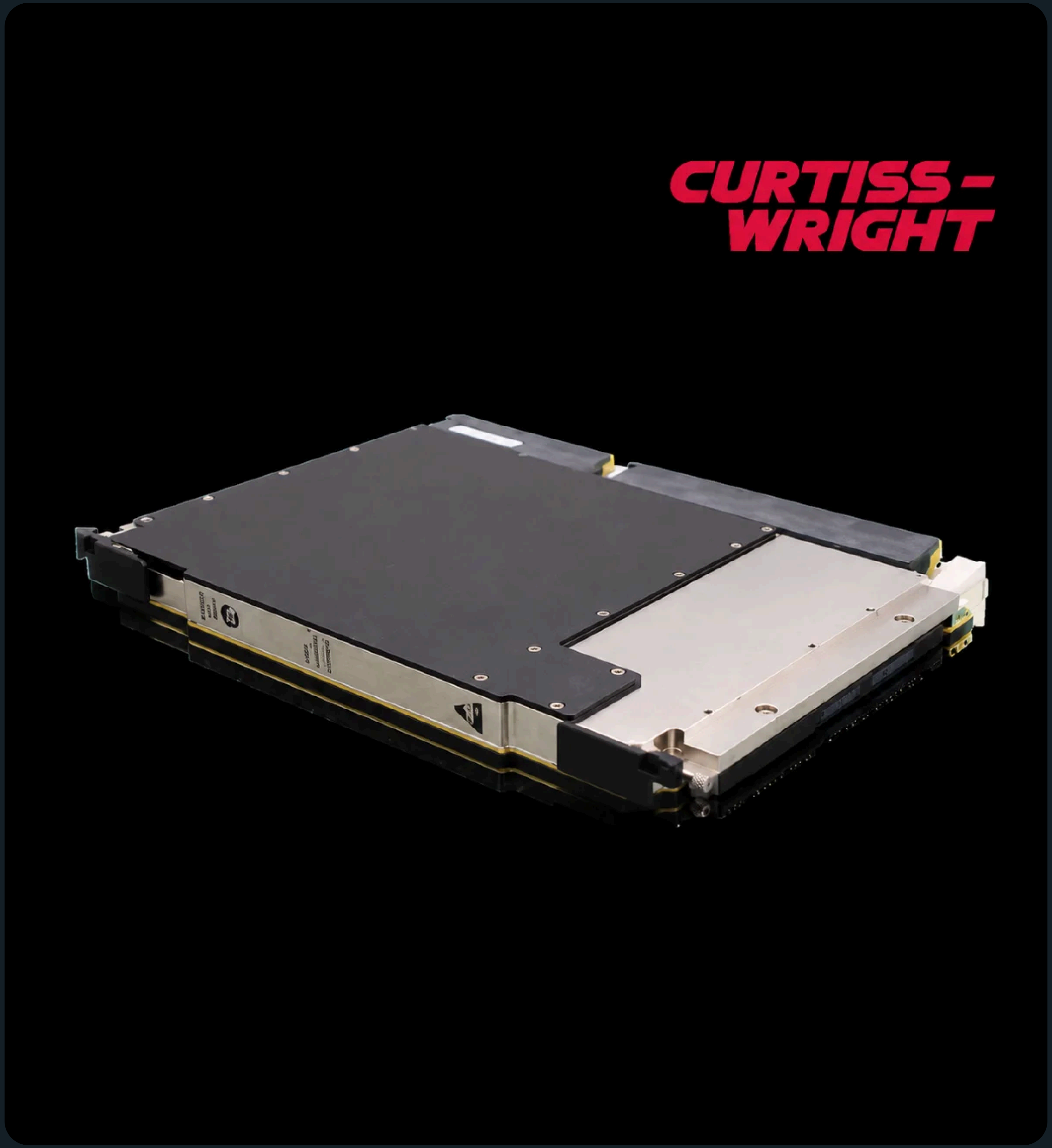
Co-op · Graduating April 2027 · Ottawa, ON

ABOUT ME

I love innovating. Building is my passion, and understanding is my method. If I see something that could be better, I work on making it better.

I am competitive, and I commit fully to whatever I take on. Right now that is a six-axis robot arm, teaching myself the piano, and training for my next fight, with travelling and scuba diving filling whatever time I have left.

Muay Thai | Scuba Diving | Volleyball | Travel | Reading



WORK CURTISS-WRIGHT · 2025

Liquid-Cooled Chassis Redesign

I reworked the liquid cooling geometry of a ruggedized electronics part to see if cheaper machining methods could hold the same thermal performance.

MY ROLE

Redrew the cooling paths in CAD around cheaper machining processes, then ran Flotherm simulations to compare the new geometry against the original.

OUTCOME

A manufacturable, lower-cost layout the team could build against, alongside thermal and environmental testing on nine conduction-cooled products.

Flotherm · Thermal design · CAD

WORK CANMETENERGY · RENEWABLE HEAT & POWER · 2025

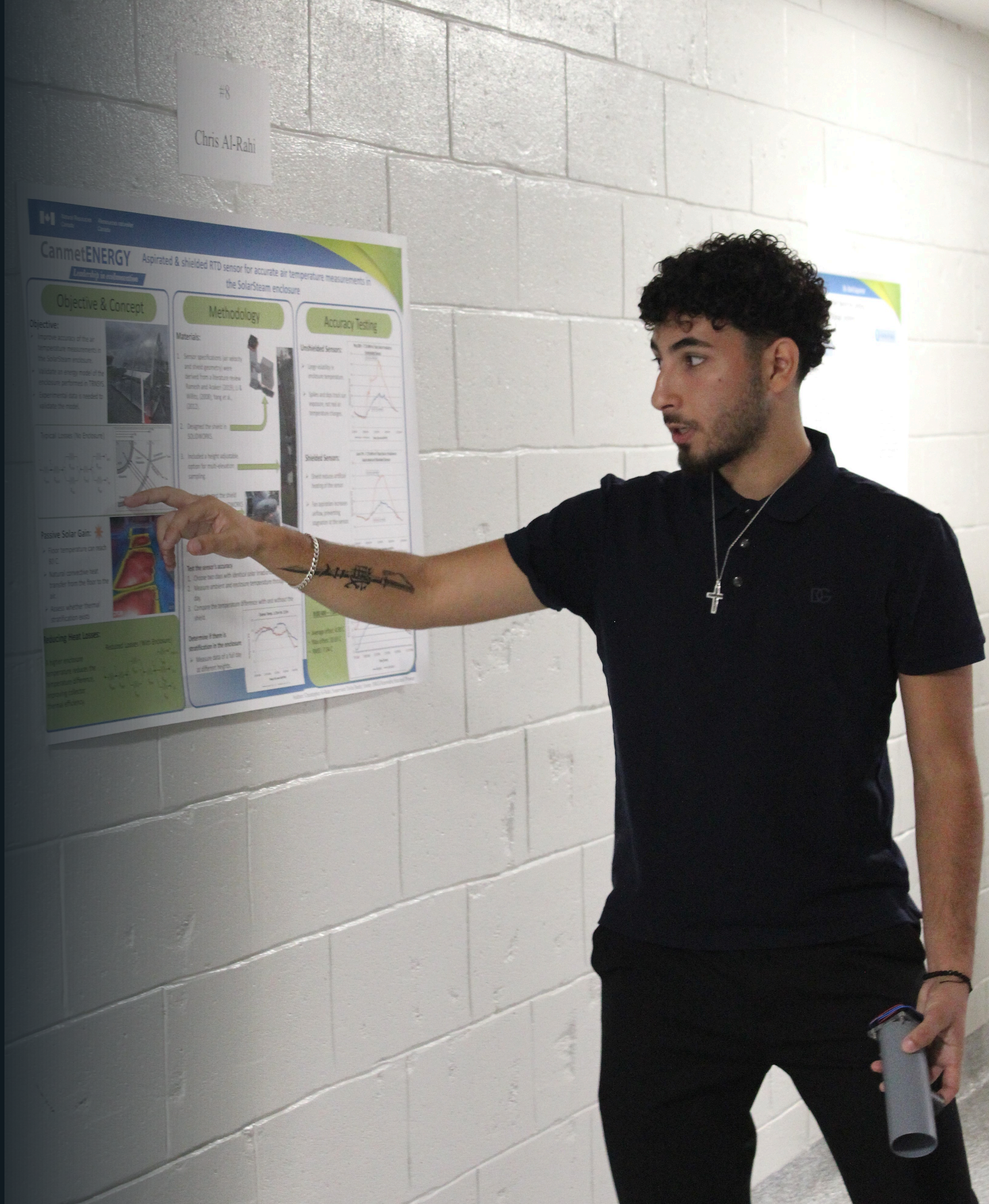
Aspirated & Shielded RTD Sensor

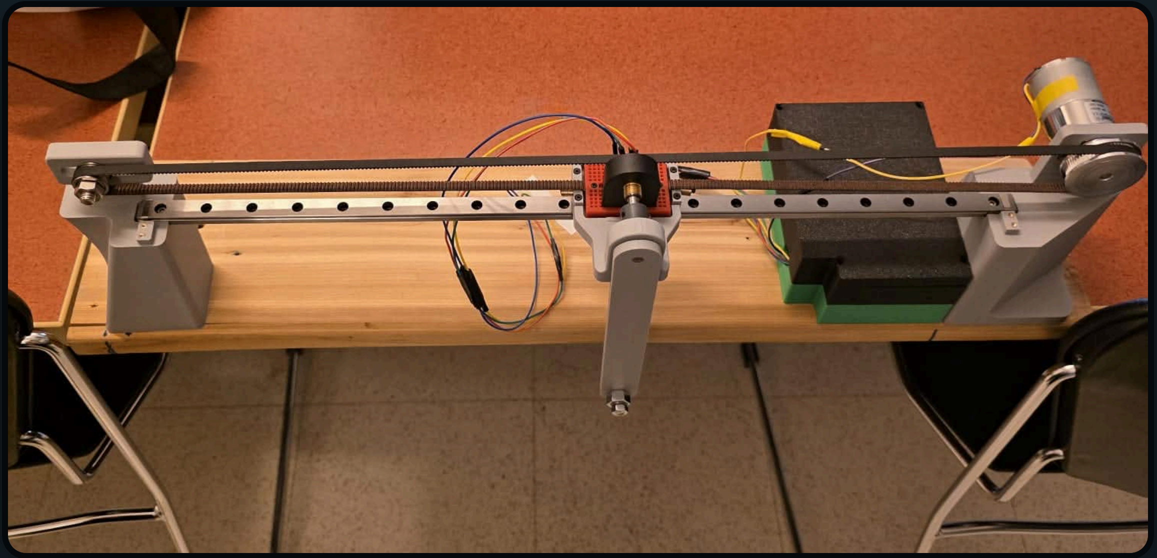
Air temperature readings inside the SolarSteam enclosure were being thrown off by radiation, so I built a shielded sensor that slides up and down to sample at any height.

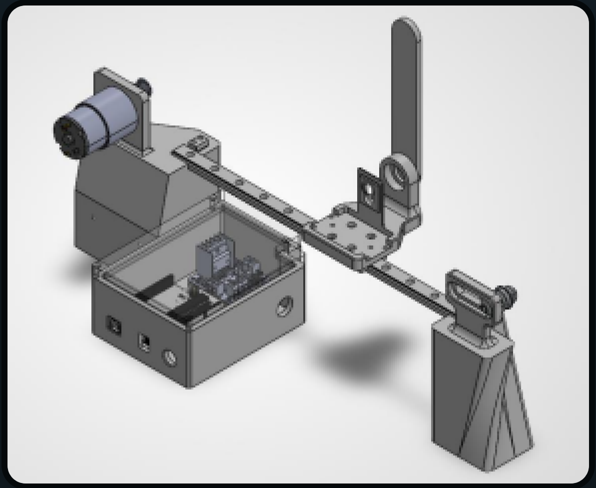
MY ROLE — Designed the shield in SolidWorks, sized the aspiration from the literature, and ran the accuracy testing myself.

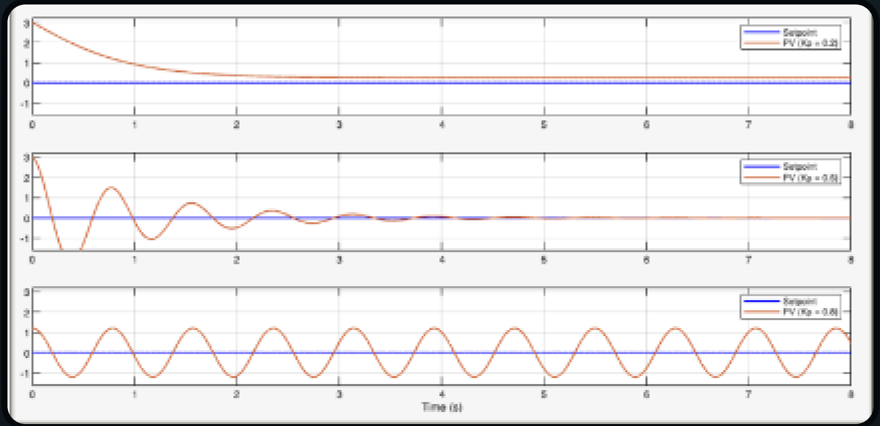
OUTCOME — Live temperature accuracy improved by more than 30%, and I presented the work as a research poster.

SolidWorks · Instrumentation · Field testing









SCHOOL CONTROL SYSTEMS

Inverted Pendulum

A belt-driven cart on a 0.5 m rail that balances a rigid pendulum upright, with a rotary encoder reading the angle and nothing telling it where the cart is.

MY ROLE Designed every 3D model on the rig, Sourced all parts, optimized the electronics layout, and helped tune the PID controller.

OUTCOME A closed-loop system that holds the pendulum upright and recovers from disturbances.

SolidWorks · Arduino (C++) · PID control · 3D printing

SCHOOL MACHINE DESIGN

Two-Stage Gear Reducer

A 9:1 reducer for an industrial conveyor drive,
taking 6 hp at 400 rpm down to 45 rpm.

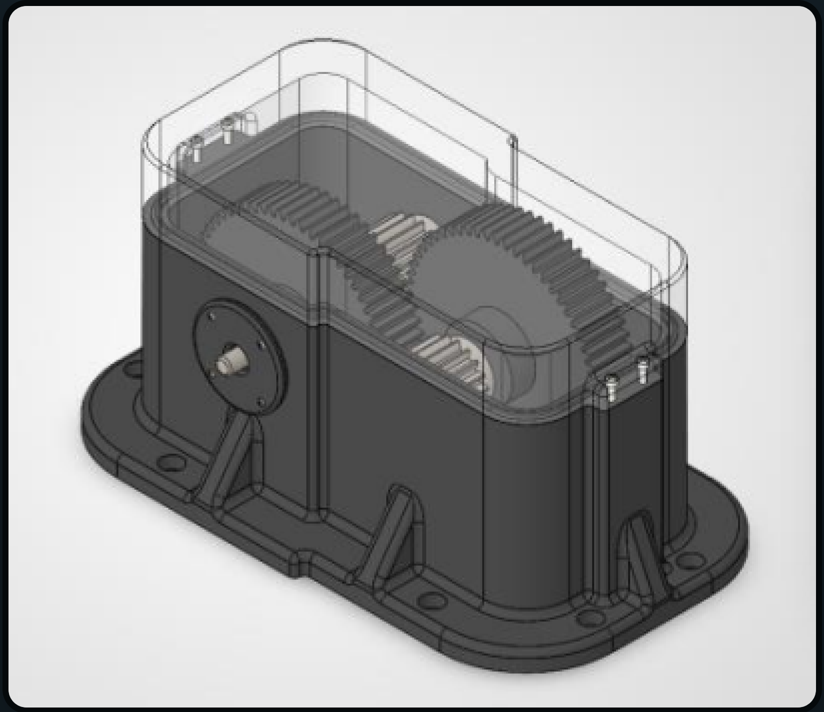
MY ROLE

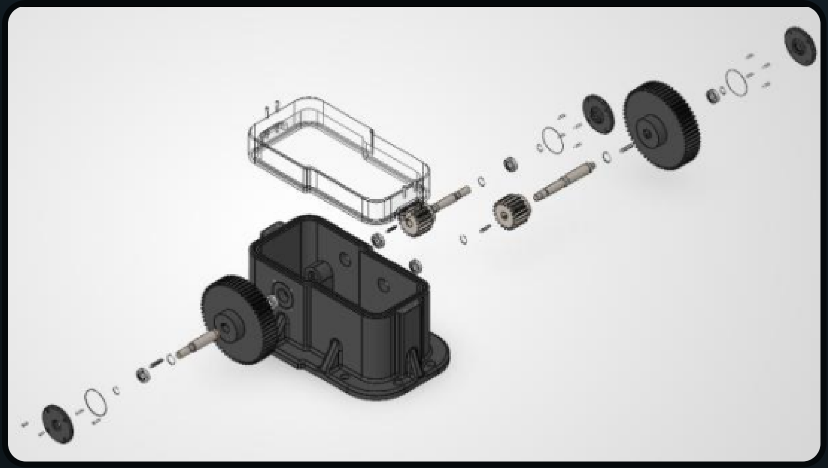
Sized the two-stage gear reduction and ran the bending and moment failure analysis. Designed every part in the assembly except the shaft.

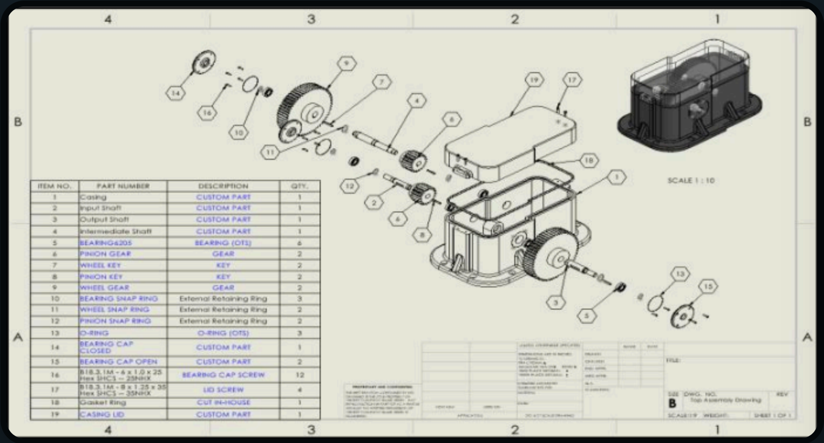
OUTCOME

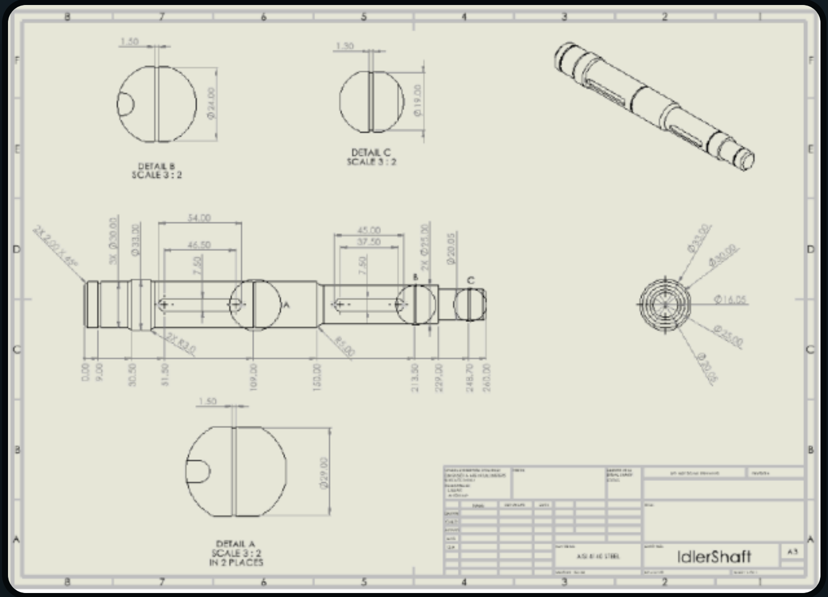
A sealed unit validated for a five-year service life, delivered with a full drawing package.

SolidWorks · Gear design · Failure analysis · Engineering drawings









SCHOOL MCG 2101

Turbine Blade Mold Polisher

A pneumatic gantry that polishes 9-metre wind turbine blade molds automatically, replacing a job that used to be slow and hard on people's arms.

MY ROLE

Designed the mechanical layout and motion system: the gantry, the yoke, and the spring-connected plates that follow the curve.

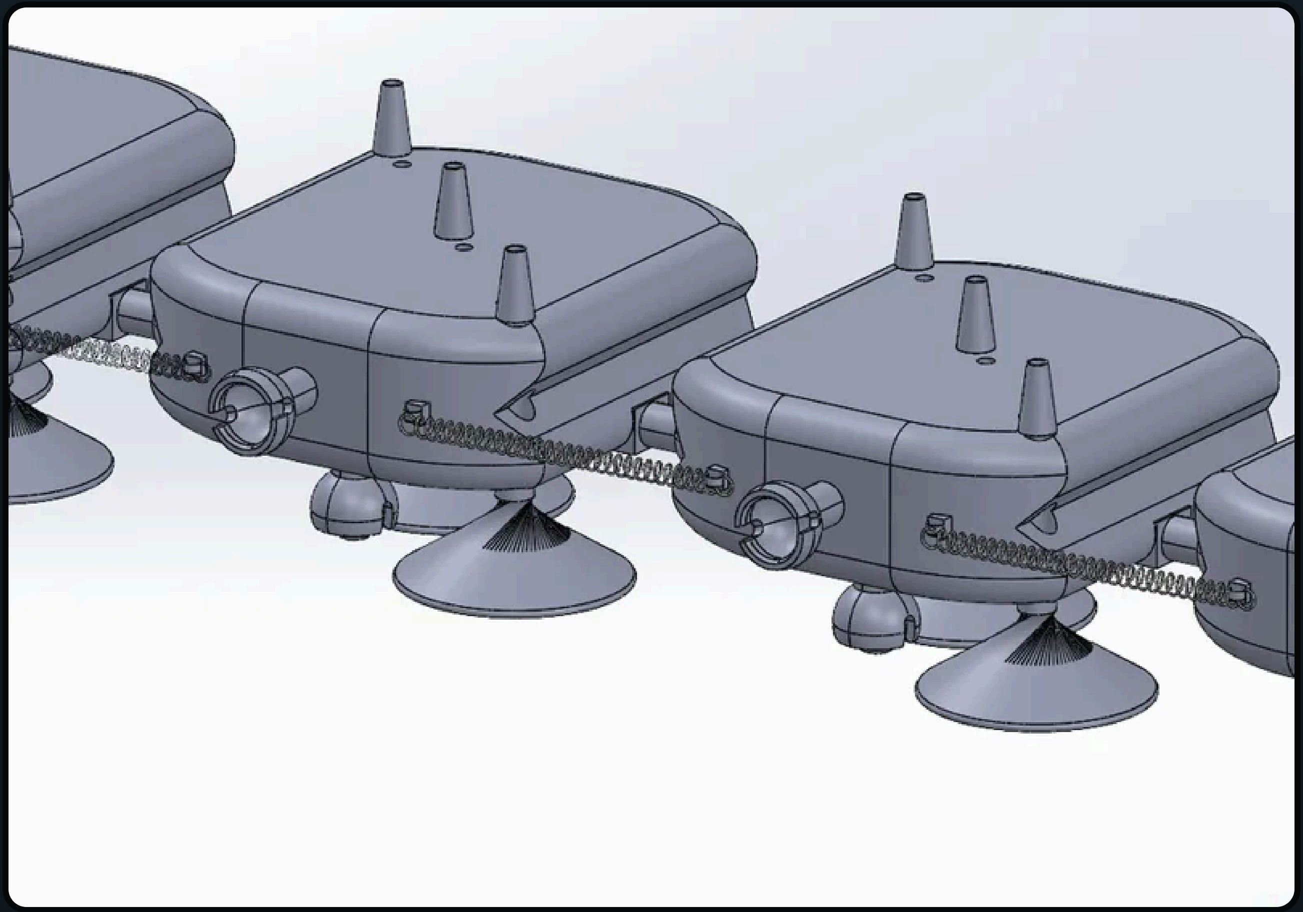
OUTCOME

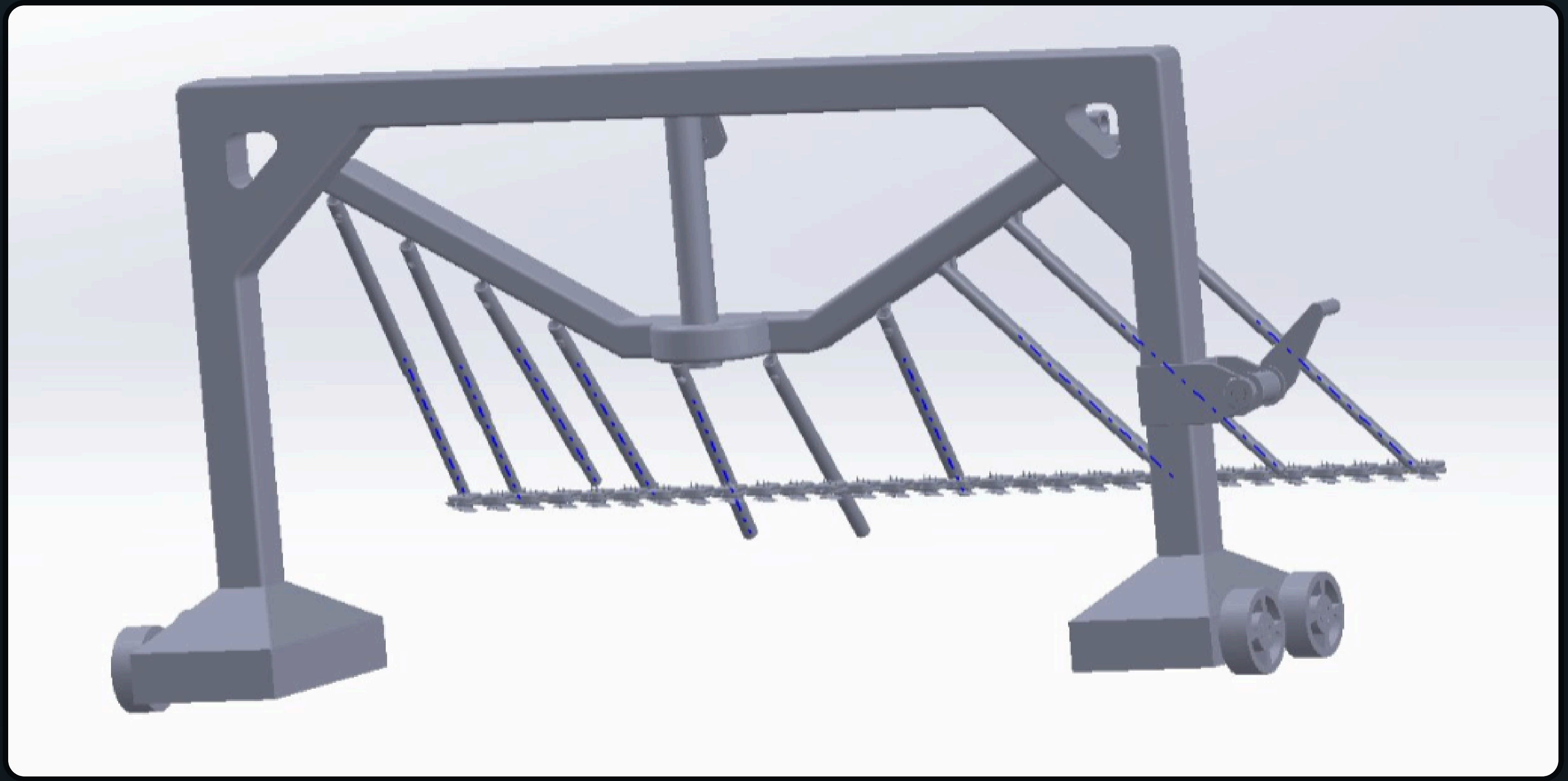
A complete CAD assembly, a full drawing package, and an interactive 3D model that holds constant pressure across the surface.

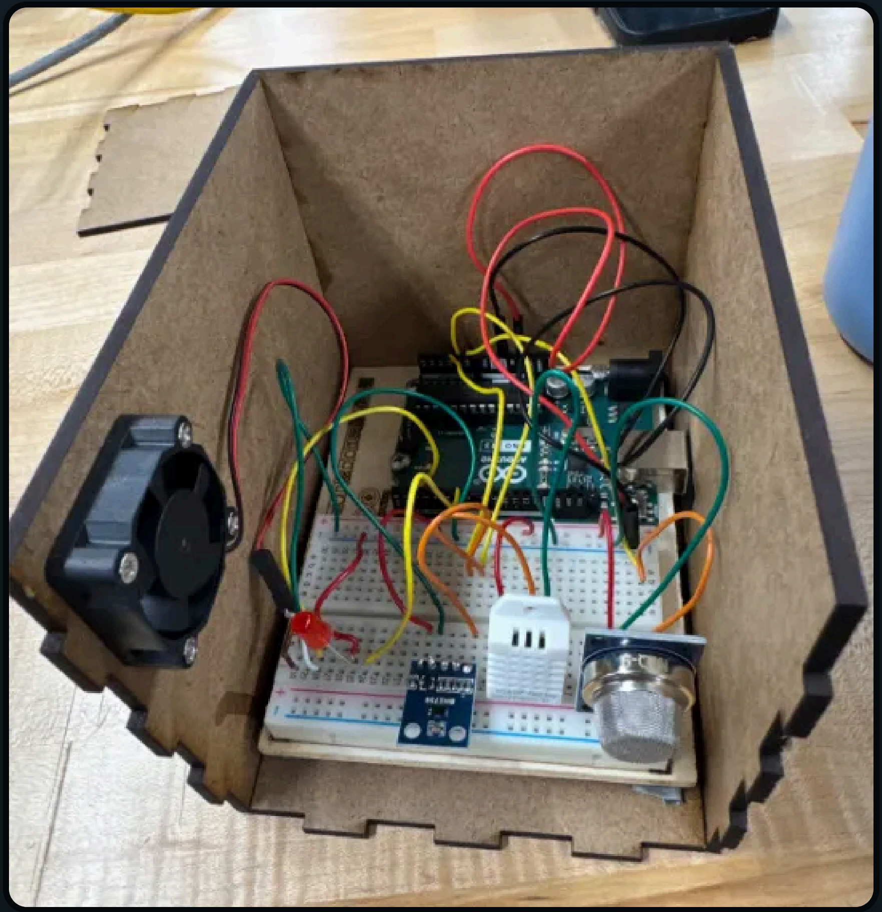
SolidWorks

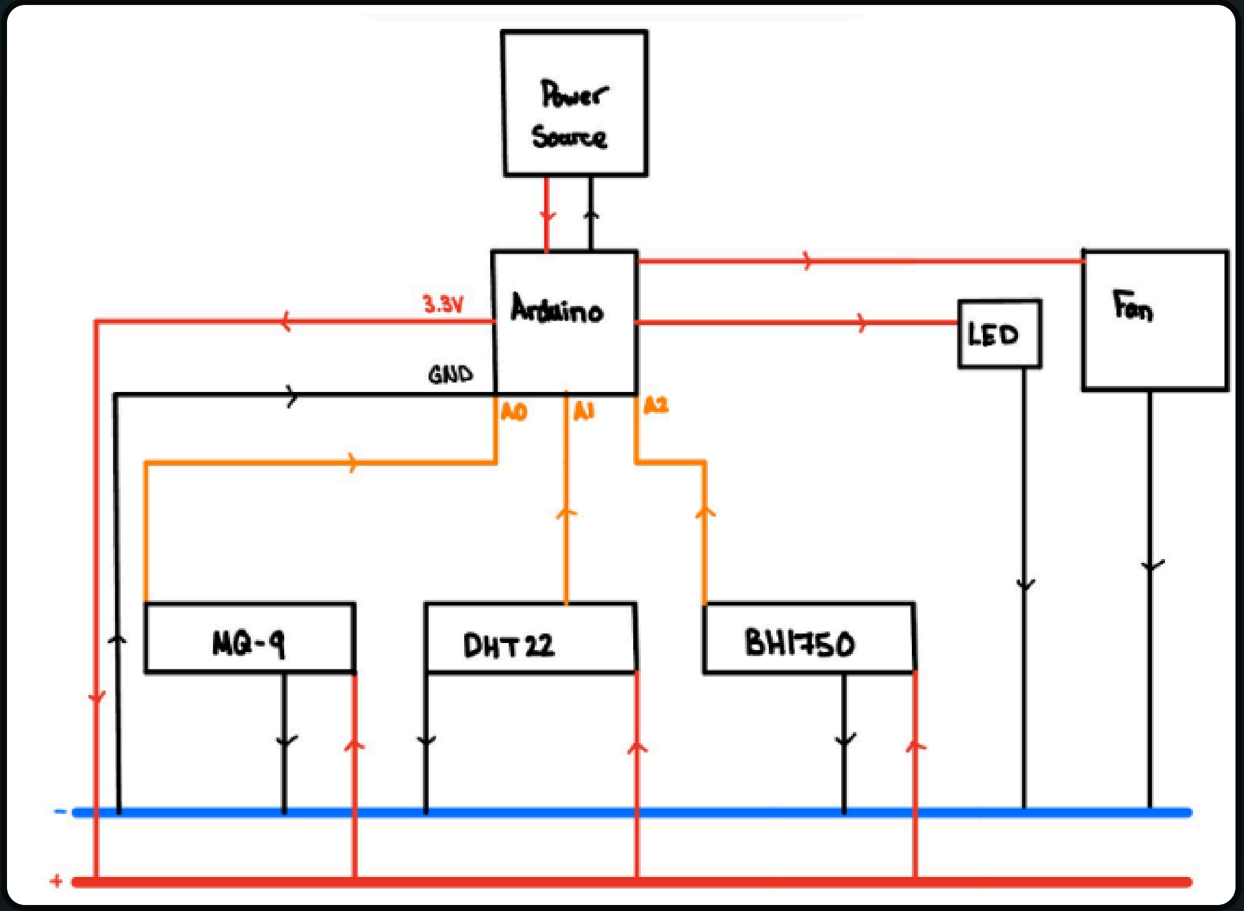
Engineering drawings

Pneumatic actuation









SCHOOL

EMBEDDED SYSTEMS · ELG 3336

Greenhouse Climate Controller

An automated setup my partner and I built to watch temperature, humidity, light, and air quality, and correct the environment on its own.

MY ROLE

We designed it together and I coded the whole thing: reading the sensors, processing the data, switching the fan and lights.

OUTCOME

A low-cost, modular controller that held the environment stable with nobody touching it.

Arduino (C++)

DHT22 / BH1750 / MQ-9

Circuit design

SCHOOL BEYOND THE PALE BREWING

Beer Goggles

A gravity and temperature monitoring system that reads straight from the fermentation tank, so the brewers don't have to sample by hand.

MY ROLE Led our team of three, handled mechanical integration, sensor calibration, and the Arduino code tying it together.

OUTCOME Read specific gravity within ±0.1 across three rounds of testing, handed off with a full user manual.

Arduino (C++) · Load cell + ultrasonic · SolidWorks · 3D printing

THANKS FOR LOOKING

Let's build something

If any of this caught your interest, or you have a project you want a hand with, reach out. I'm always up for collaborating.

chris.alrahi@gmail.com

LinkedIn

My Website

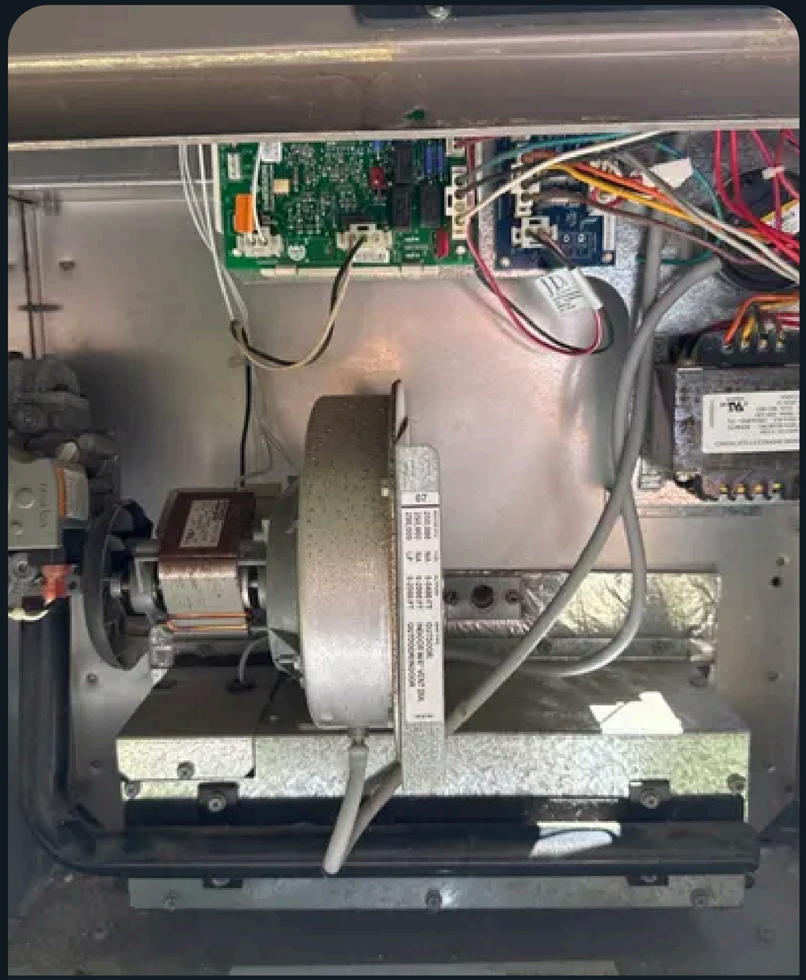
HONORABLE MENTIONS

Builds & Repairs

Things I've built or fixed on my own time, a jack of all and a master of builds.

Car Fender Repair

Filled and sanded the cracked panel, matched the paint, refit it myself.

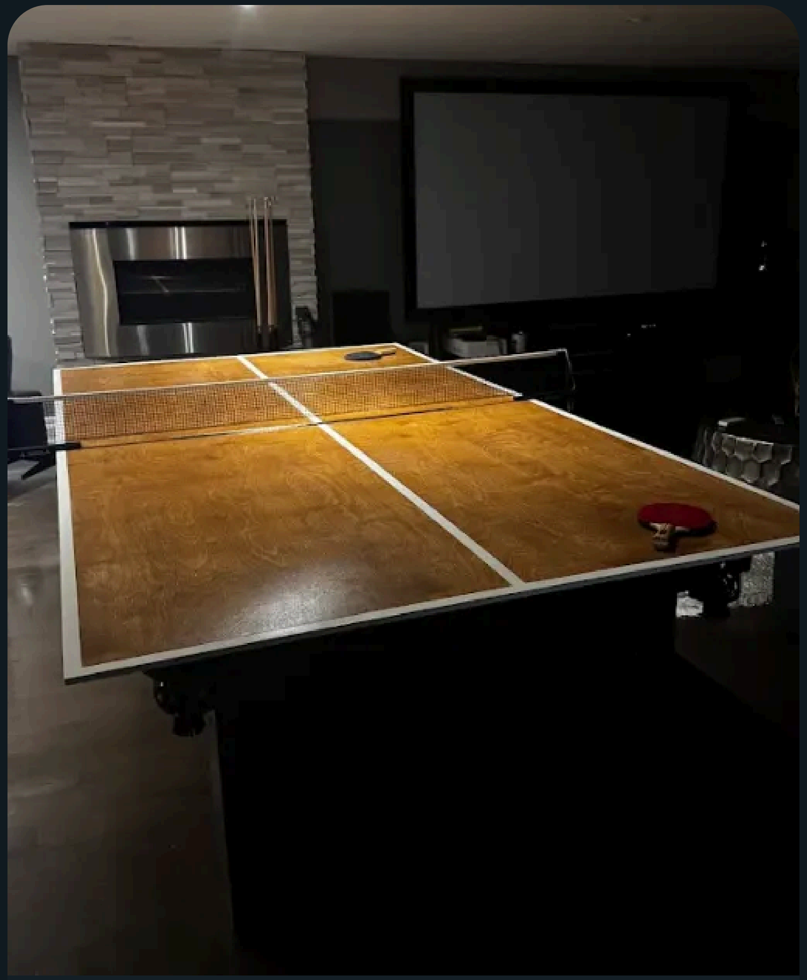


Pool Heater

Traced the fault down to the board and blown components, sourced, and repaired it.

Hot-Tub Heater

Found the electrical fault, sourced an element, put it back in the piping.



Ping-Pong Top

A fold-away top that turns our pool table into a ping pong table in seconds.